

Lab 3_report.rmd

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Lab 3: Reproducible Research in R

Checkpoint 4: How do we interpret this summary?

The summary table for the ANOVA shows the degrees of freedom, summary of squares, mean of squares (variance), F value (ratio of between-group to within-group variance) and the p values associated with each comparison. None of the 3 p values generated meet the significance threshold ($p = 0.05$), indicating that the difference between the means is not statistically significant. In this scenario, the effects of both population of origin and temperature treatment on the growth rate are not statistically significant.

Checkpoint 5. What did we just do? Do you think this approach is easily editable and reproducible?

In order to compare the effect of the temperature treatments and population of origin on the growth rate, we created a boxplot showing the growth rate as a function of both temperature treatments and both populations of origin. To determine whether the effects were statistically significant, we ran an ANOVA to calculate the difference between the means of each category. I do think the approach is editable and reproducible; what's currently lacking in the workflow would be a Readme to explain the full process.

```
##Code block for "EditDataframe.R"

##Import data

temp.df <- read.csv("data/tempExperiment-raw.csv", header = T)
temp.df

head(temp.df)

print(temp.df)

#separate temperature treatment from population using strsplit
a <- strsplit(as.character(temp.df$temp), split = '-')
#use unlist and matrix functions to rearranging the output from strsplit
newvar <- matrix(unlist(a), ncol = 2, byrow = TRUE)
head(newvar)
#make a new dataframe with just temp and a new column for pop
df.t2 <- temp.df #set the new dataframe to the original so we can fix it
df.t2$temp <- newvar[,1]
df.t2$pop <- newvar[,2]
head(df.t2)

#use sub to give temp only numeric values and pop only levels pop 1 and pop 2, watch the space at the e
```

```

df.t2$temp <- sub('ten', '10', df.t2$temp)
df.t2$temp <- sub('twenty', '20', df.t2$temp)
df.t2$pop <- sub('population ', 'pop', df.t2$pop)
str(df.t2)
#Make sure all objects are correct flavor using factor() or as.numeric
df.t2$temp <- factor(df.t2$temp)
df.t2$pop <- factor(df.t2$pop)
str(df.t2)

#Export a new edited csv
write.csv(df.t2, 'data/tempExperiment_v2.csv')

```

```
##Code chunk for Analysis.R
```

```

#start up from last step in EditDataframe.R, which was to export a CSV of our cleaned up data.
#Clear the environment
rm(list=ls(all=TRUE))
#Import the cleaned data as a new df
df.t <- read.csv('data/tempExperiment_v2.csv')
#Get an overview
str(df.t)

```

```

#R thinks temp is a nubmer but really it's a low or high temp category, so change temp to a factor
df.t$temp <- factor(df.t$temp)

```

```

##want to analyze how temperature treatment, population of origin, and size affect growth rate

```

```

#Plot the data - boxplot of variable growth rate as a function of (~) temp and pop. + tells it to combi
boxplot(df.t$growthRate~df.t$temp+df.t$pop)

```

```

#add axis labels

```

```

boxplot(df.t$growthRate~df.t$temp+df.t$pop,
  names = c('10', '20', '10', '20'), #temperature label
  at = c(1,2,4,5), #space out boxes according to population - I don't get this
  ylab = 'Growth rate mm/day',
  xlab = ''
)

```

```

#use mtext to add text in the margin - not an axis label b/c you want to space it with the boxplot. idk

```

```

mtext('Pop 1', side = 1, at = 1.5, line = 3)
mtext('Pop 2', side = 1, at = 4.5, line = 3)

```

```

#Now save it to PDF in results folder

```

```

pdf('results/MyBoxplot.pdf', width = 5, height = 5)
##Use code from above
boxplot(df.t$growthRate~df.t$temp+df.t$pop,
  names = c('10', '20', '10', '20'), #temperature label
  at = c(1,2,4,5), #space out boxes according to population - I don't get this

```

```

    ylab = 'Growth rate mm/day',
    xlab = ''
)

#use mtext to add text in the margin - not an axis label b/c you want to space it with the boxplot. idk

mtext('Pop 1', side = 1, at = 1.5, line = 3)
mtext('Pop 2', side = 1, at = 4.5, line = 3)

dev.off()

##Now find out if effect of pop or effect of temp is significant in analysis. Use ANOVA.
# The '*' symbol is for multiplication and means we want to evaluate an interaction, in this case evaluate

m1 <- aov(df.t$growthRate ~ df.t$temp * df.t$pop)
m1.summary <- summary(m1)
m1.summary

##save the summary table as an R object so you can call it later
saveRDS(m1.summary, 'results/m1.summary.rds')

##Code chunk for "driver.R"

##Run to call the source code for analysis

source('src/EditDataframe.R')

source('src/Analysis.R')

```

*Used Google AI to debug issues with knitr & forcing the working directory for the R markdown file to be my main class folder instead of the 'reports' folder where the markdown file is stored.